



Microstructure and Mechanical Properties of Electron Beam Melted Nb–W and Nb–W–Zr Alloys

M. Sankar¹ · Shishir Chaudhary¹ · V. V. Satya Prasad¹ · B. Gopalakrishna¹

Received: 6 May 2023 / Accepted: 18 September 2023 / Published online: 12 October 2023
© The Indian Institute of Metals - IIM 2023

Abstract The present work deals with microstructure evolution and room temperature mechanical properties of electron beam melted Nb–10 wt.% W (Nb–W) and Nb–10 wt.% W–2.5 wt.% Zr (Nb–W–Zr) alloys. The consumable electrodes were prepared through mechanical mixing, cold isostatic pressing and vacuum sintering using the raw materials in the form of powders. These electrodes were used for the preparation of ingots through electron beam melting. The ingots were forged at 1250 °C; subsequently, they were subjected to warm rolling at 1000 °C and cold rolling at 25 °C with an intermediate annealing. The alloys were characterised by optical microscope, scanning electron microscopy, electron backscattered diffraction and room temperature tensile test. The results show that both alloys exhibit a single-phase microstructure. The Nb–W–Zr alloy displays a higher tensile strength as compared to the Nb–W alloy. The Nb–W alloy under warm rolled and annealed condition shows a sharp yield point during tensile test, while such a sharp yield point is not observed for Nb–W–Zr alloy. The fracture surfaces of tensile tested samples reveal that Nb–W alloy shows a completely transgranular cleavage fracture. A mixed mode of fracture consisting of both dimple and cleavage is observed for Nb–W–Zr alloy.

Keywords Niobium alloy · Electron beam melting · Hot forging · ZrO₂ particle

1 Introduction

Niobium and its alloys are one of the most promising high temperature materials for aerospace, defence and nuclear applications owing to their high melting point, good strength at high temperature and relatively low density along with good workability and weldability [1–3]. The envisaged applications of these alloys are beyond the application temperature of commonly used high temperature nickel base superalloys (1000 °C). Unalloyed niobium generally exhibits a low strength at elevated temperature and is not suitable for any high temperature applications. Hence, niobium is alloyed with a variety of alloying elements such as Mo, W, Ti, Zr and Hf to enhance its high temperature strength [4, 5]. The alloying elements such as Mo and W have been added to Nb to improve their high temperature strength due to solid solution strengthening, while elements such as Hf and Zr have been added both for solid solution strengthening as well as for dispersion strengthening [5, 6]. Among the niobium-based alloys, C-103 (Nb–10Hf–1Ti) and Cb-752 (Nb–10W–2.5Zr) are ideally suited for fabrication of an actively cooled combustor chamber for hypersonic vehicles and thrust chamber of rocket because of their combination of high temperature strength and room temperature ductility [4]. However, the strength of Cb-752 alloy is better than that of C-103 alloy at all temperatures. The strengthening effect of Cb-752 alloy comes from both solid solution strengthening as well as dispersion strengthening.

The conventional melting and casting routes are generally not suitable for preparation of niobium-based alloys because their melting temperatures are above the working temperatures of most of the common refractories. Further, the mechanical properties of Nb-based alloys largely depend on the presences of interstitial impurities such as oxygen, nitrogen, hydrogen and carbon. The Nb alloys become

✉ M. Sankar
ulundurmasankar@gmail.com

¹ Defence Metallurgical Research Laboratory,
Kanchanbagh Hyderabad-500058, Telangana, India

brittle if these impurities are not controlled properly and kept below the tolerable limit [7, 8]. Hence, special melting techniques such as electron beam melting (EBM) and vacuum arc remelting (VAR) are commonly employed processes for preparation of Nb and its alloys [9]. The EBM process is primarily employed for purification and preparation of high-purity Nb ingots. It is also used for the preparation of Nb-based alloys having alloying elements with similar vapour pressure as Nb [10]. The high temperature and high vacuum encountered during EBM process result in production of high-purity ingot with low levels of interstitial impurities, whereas the VAR process is widely employed for the preparation of Nb-based alloys containing alloying elements having high-purity and high vapour pressure [11]. The alloy Cb–752 contains both low vapour pressure element W and high vapour pressure element Zr apart from differences in melting temperatures and densities as compared to Nb. Therefore, it is a real challenge for a metallurgist to prepare this alloy. Till date the information available in the literature pertaining to melting, processing and fabrication technologies of Nb–10 W–2.5Zr alloy is scanty and inadequate. Hence, through the present work, an attempt has been made to prepare Nb–10 wt.% W–2.5 wt.% Zr alloys through EB melting technique using the raw materials in the form of elemental powders. Further, we process them under different processing conditions to make into sheets and compare the microstructure and mechanical properties. An attempt has also been made to prepare and process Nb–10 wt.% W alloy for the purpose of comparison.

2 Experimental

The raw materials used in the present work are Nb (99.5%), W (99.9%) and Zr (99.0%) powders. Nb–W and Nb–W–Zr alloy electrodes with a nominal composition of Nb–9% W and Nb–8% W–12%Zr were prepared through powder metallurgy route using mechanical mixing, cold isostatic pressing and vacuum sintering with an aimed average chemical composition of Nb–10% W and Nb–10% W–2.5% Zr after EB melting. The chemical composition of the alloys is expressed in weight per cent throughout the text unless otherwise stated. Hereafter, Nb–10% W alloy is referred as Nb–W and Nb–10% W–2.5% Zr alloy is referred as Nb–W–Zr. The electrodes were drip melted in a water cooled copper crucible using a 60 kW EBM furnace. A single melting was used for preparation of Nb–W alloy ingot, while double melting was used for preparation of Nb–W–Zr alloy ingot. A power input of 45 kW and a mould of diameter 60 mm were used for melting of both the alloys.

The 60 mm diameter ingots were vacuum annealed at 1400 °C for 24 h for homogenization. Then, they were machined to 56 mm diameter and applied coating with

composition of Si–20% Cr–20% Fe. The green coating was diffusion annealed at 1400 °C under 1×10^{-5} m bar vacuum for 2 h. The coated ingots were heated to 1250 °C and forged to achieve a final thickness of 26 mm using a pneumatic forge hammer of one ton capacity. The hot forged plates were cut, machined to 18 mm thick and vacuum annealed at 1400 °C for 12 h. They were then warm rolled at 1000 °C with a commercially available deltaglaze coating to get 9 mm thick plates, which were again annealed at 1400 °C for 4 h. They were again rolled at room temperature to get sheets with a final thickness of 3 mm and finally annealed at 1350 °C for 1 h. All metallic impurities present in the alloys were analysed by inductively coupled plasma optical emission spectroscopy (ICPOES). A LECO carbon and sulphur analyser was used for analysis of carbon, whereas a LECO hydrogen determinator was used for analysis of hydrogen. The analysis of oxygen and nitrogen were carried out using a LECO oxygen–nitrogen gas analyser.

Standard metallographic techniques were used for preparation of samples for microstructural observations. The samples were etched with an etchant consisting of 15 ml H₂O, 30 ml Hf, 30 ml HF, 15 ml H₂SO₄ and 10 ml HNO₃. The microstructure of the alloys was examined using optical microscope, scanning electron microscope and electron backscattered diffraction. The mechanical properties of the alloys at room temperature were evaluated by tensile test. Tensile samples were prepared according to ASTM E8M standard. Tensile tests were conducted using an Instron Universal testing machine at an initial strain rate of 1×10^{-3} per second. The fracture surfaces of tensile tested samples were examined under SEM.

3 Results and Discussion

3.1 Characterisation of as-Cast Alloys

The as-cast macrostructure of Nb–W and Nb–W–Zr alloys is shown in Fig. 1. It can be seen that Nb–W alloy shows a three distinct regions of grain structure such as chilled zone, columnar zone and equiaxed zone, while Nb–W–Zr alloy does not exhibit such distinct regions of grain structure. Instead, the Nb–W–Zr alloy displays a relatively fine and uniform macrostructure. It is known from solidification of alloys that, when a liquid metal contacts the mould surface the chilled zone is formed due to the fast cooling of molten metal. The grain structure of chilled zones is fine, equiaxed and has a random crystallographic orientation. The nucleation of grains in the chilled zone is attributed to undercooling effect that occurs when the molten liquid first makes the contact with a cold mould wall. As the solidification continues towards the centre of the casting, the undercooling is reduced, and the grains grow in an elongated manner. As

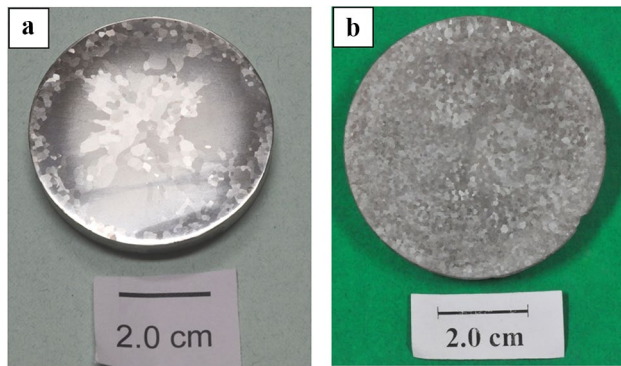


Fig. 1 Photographs showing macrostructure of transverse section of EB melted (a) Nb–W alloy and (b) Nb–W–Zr alloy ingots

As a result, the columnar grains are obtained. As the solidification progresses further, the molten pool at the centre of the casting results in the formation of uniform grains structure. It is known that the centre equiaxed grains nucleate due to a significant amount of supercooling exists due to the solute segregation. The supercooling of alloy due to solute segregation is called the constitutional supercooling [12]. It generally arises from the fact that the solute segregation lowers the local liquidus temperature of alloy. The relatively fine macrostructure observed for Nb–W–Zr alloy is attributed to the presence of Zr in the alloy. It is generally known that Zr is a reactive metal, and it reacts with a residual impurities particularly oxygen present in the molten pool during solidification. As a result ZrO_2 particles are formed in Nb–W–Zr alloy before the commencement of solidification. These ZrO_2 particles act as a heterogeneous nucleation sites during solidification and aid for refinement of grain structure in Nb–W–Zr alloy. Moreover, the melting point of ZrO_2 (2780 °C) is high as compared to melting temperature of Nb–W–Zr alloy (2450 °C) [13].

It is generally accepted that ZrO_2 is highly stable once it is formed because of its large negative formation of free energy (ΔG) as compared to the free energy formation of

niobium and tungsten oxides. Besides, it is known that ZrO_2 generally exist in a three allotropic forms depending upon the temperatures [14]. It has a cubic crystal structure form at high temperature (> 2380 °C) with a lattice parameters of $a=b=c=5.09 \text{ \AA}$. It transforms to a tetragonal form of structure at 2380 °C and below with a lattice parameters of $a=b=3.57 \text{ \AA}$, $c=5.18 \text{ \AA}$. Later, it transforms to a monoclinic crystal structure at 1205 °C and below with a lattice parameters of $a=5.15 \text{ \AA}$, $b=5.21 \text{ \AA}$ and $c=5.31 \text{ \AA}$. However, the phase transformation from a tetragonal form of ZrO_2 into a monoclinic form of ZrO_2 is reported to be sluggish in nature [15]. It may be inferred that the cubic form of ZrO_2 particles is likely to act as a heterogeneous nucleation sites for Nb–W–Zr alloy during solidification. Moreover, the structural compatibility is expected between Nb and ZrO_2 particles at high temperature because of both exist in BCC structure. As a result, the molten liquid easily wets the ZrO_2 particles. Thus, the cubic ZrO_2 assists for increasing the nucleation rate to obtain a fine grain structure in Nb–W–Zr alloy.

The chemical composition of Nb–W and Nb–W–Zr alloys is shown in Table 1. It can be noted that Nb–W alloy is melted once, while Nb–W–Zr alloy is melted twice in order to get the required chemical composition in the ingots as the initial oxygen content is not same for both alloys. It is clearly seen from the Table 1 that the initial concentration of oxygen in Nb–W–Zr alloy is 0.65 wt.%, and in Nb–W alloy is 0.25 wt.%. The removal of a large quantity of oxygen from Nb–W–Zr is very difficult as compared to Nb–W alloy. Hence, Nb–W–Zr alloy is initially melted at lower melt rate in order to remove maximum oxygen content, and subsequently, it is melted at a higher melt rate in order to reduce the losses of Zr during melting. It is known that the lower melt rates allow a longer residence time for the molten pool in the liquid state and enables better removal of oxygen during melting of the alloy. It is obvious from Table 1 that the oxygen content is reduced to very low levels (well below the specified upper limits) after double melting of alloy. Besides the other interstitial impurities such as C, N

Table 1 Chemical composition of Nb–W and Nb–W–Zr alloys before and after EB melting

Elements	Chemical composition (wt.%)					
	Nb–W		Nb–W–Zr			
	Alloy Electrode	After Melting	Alloy Electrode	After 1 st Melting	After 2 nd Melting	
C	0.045 ± 0.005	0.005 ± 0.001	0.06 ± 0.005	0.007 ± 0.002	0.006 ± 0.002	
H	0.09 ± 0.002	0.0008 ± 0.0001	0.08 ± 0.002	0.0007 ± 0.0001	0.0002 ± 0.0001	
N	0.04 ± 0.003	0.05 ± 0.003	0.04 ± 0.003	0.0092 ± 0.002	0.0035 ± 0.003	
O	0.25 ± 0.005	0.018 ± 0.04	0.65 ± 0.08	0.0478 ± 0.05	0.013 ± 0.03	
W	9.0 ± 0.3	10.6 ± 0.24	8.5 ± 0.2	10.28 ± 0.1	10.82 ± 0.09	
Zr	–	–	12.5 ± 0.22	4.8 ± 0.04	2.43 ± 0.05	
Nb	Balance	Balance	Balance	Balance	Balance	

Fig. 2 Optical microstructure of alloys processed under warm rolled and annealed conditions: (a) Nb–W alloy and (b) Nb–W–Zr alloy

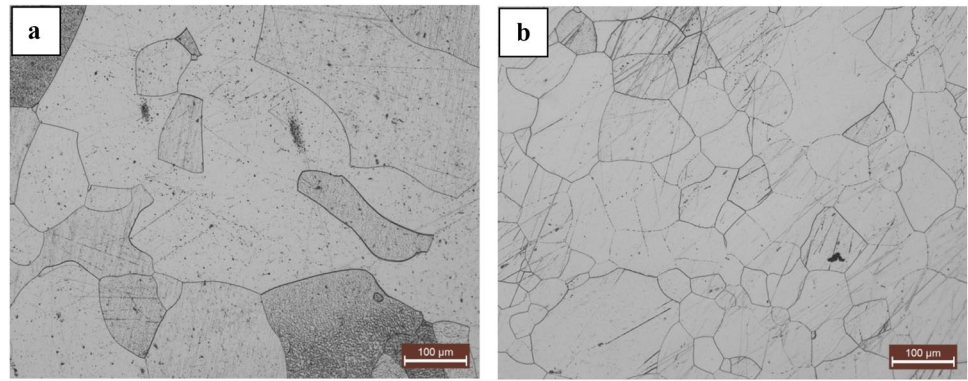
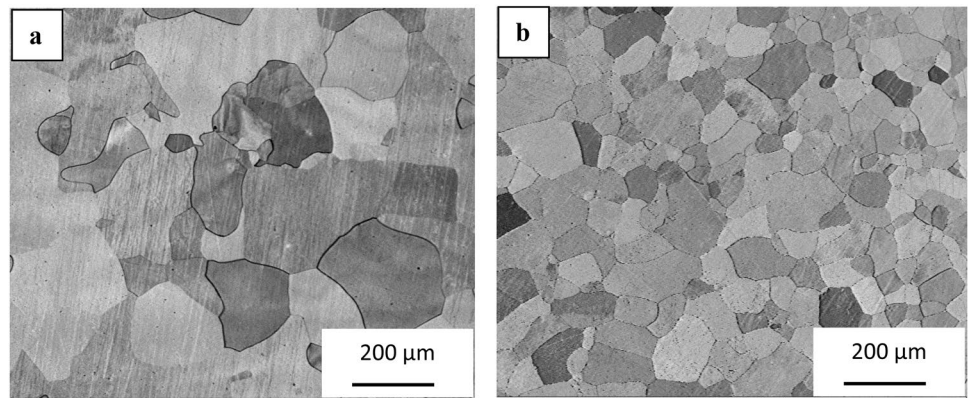


Fig. 3 SEM micrographs of alloys processed under warm rolled and annealed conditions: (a) Nb–W alloy and (b) Nb–W–Zr alloy



and H also get refined to a very low level during EB melting. It is generally accepted that the removal of gaseous impurities (O, H, N) dissolved in Nb is governed by the reversible equilibrium represented by Sievert's law of dissolution of gases in the metals [16]. The nitrogen is removed as nitrogen during melting. While the oxygen is reported to be removed either as carbon monoxide through the reaction of oxygen and carbon ($2C + O_2 = 2CO$), or as niobium sub-oxides such as NbO, NbO₂ via Nb and oxygen reaction [17, 18].

3.2 Characterisation of Thermomechanically Processed Alloys

The optical and SEM microstructures of alloys processed under warm rolling and annealing conditions are illustrated in Figs. 2 and 3, respectively. It can be seen that both alloys exhibit a single-phase microstructure with completely recrystallized grains. However, it is found that the grain sizes are different for the alloys. The grain size of Nb–W–Zr alloy is relatively fine as compared to grain size of Nb–W alloy. This may be attributed to the presence of ZrO₂ particles in Nb–W–Zr alloy. These ZrO₂ particles inhibit the grain growth during annealing of warm worked alloy. The tensile graphs of warm rolled and annealed alloys are shown in Fig. 4. It can be observed that the Nb–W alloy exhibits a sharp yield point while such a sharp yield point is not

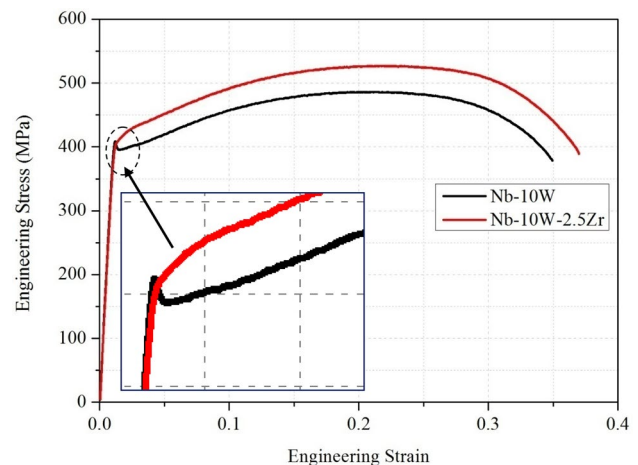


Fig. 4 Plot showing tensile curves of warm rolled and annealed Nb–W and Nb–W–Zr alloys

observed for Nb–W–Zr alloy during the tensile test. It is believed that the yield point phenomena occurring in steels during the tensile testing at low temperature is associated with the presence of interstitial impurities, particularly carbon [19]. It has also been reported for commercial grade Nb [20] and Ta [21, 22], that the occurrence of yield point phenomenon is clearly associated with presence of residual

Table 2 Tensile properties of Nb–W and Nb–W–Zr alloys processed under warm rolled and annealed condition

Alloys	Mechanical properties		
	YS (MPa)	UTS (MPa)	Elongation (%)
Nb–W	401 ± 4.5	484 ± 2	31.5 ± 1.5
Nb–W–Zr	403 ± 5	528 ± 1	35 ± 1

oxygen. Therefore, in the present study, it is believed that the presence of residual oxygen is solely responsible for the occurrence of the sharp yield point in the Nb–W alloy. In case of Nb–W–Zr alloy, the reactive element Zr scavenges the residual oxygen and forms the fine dispersoids of ZrO_2 . As a result, the matrix is free from the oxygen content. Thus, the yield point phenomenon does not take place in Nb–W–Zr alloy. The factor that is responsible for the occurrence of the yield point phenomenon in BCC refractory metals and their alloys has been reported elsewhere [23]. The tensile properties of warm rolled and annealed alloys are shown in Table 2. It is seen in Table 2 that the strength of Nb–W–Zr alloy is higher than that of Nb–W alloy. However, the ductility values are not significantly changed in the alloys. The fracture surfaces of tensile tested samples of warm rolled and annealed alloys are illustrated in Fig. 5. The Nb–W alloy displays a completely transgranular cleavage fracture, while

Nb–W–Zr alloy exhibits a mixed mode of fracture. It indicates that the fracture behaviour of alloy is changed from a completely transgranular cleavage mode to a mixed mode of fracture due to the presence of Zr in Nb–W–Zr alloy as compared to the Zr free Nb–W alloy. It is also accepted that the BCC metals and their alloys generally fail in a transgranular cleavage fracture mode if their grains sizes are very coarse [24].

The typical optical and SEM microstructure of cold rolled and annealed alloys are illustrated in Fig. 6. and Fig. 7. It can be seen that both alloys exhibit a completely recrystallized grains. As expected the grains size of Nb–W–Zr alloy is relatively fine as compared to grains size of Nb–W alloy. The tensile plots of alloys processed under cold rolling and annealing conditions are depicted in Fig. 8. It is clear from Fig. 8 that both Nb–W and Nb–W–Zr alloys show a completely smooth tensile curve as opposite to the sharp yield point observed for warm rolled and annealed Nb–W alloy. This is quite interesting and the reasons for nonoccurrence of a sharp yield point in cold rolled and annealed Nb–W alloy is yet to be explored. The tensile properties of cold rolled and annealed alloys are shown in Table 3. It can be seen that as expected the strength of Nb–W–Zr alloy is higher than that of Nb–W alloy. However, the strength of both the alloys is higher as compared to the strength of warm rolled and annealed alloys.

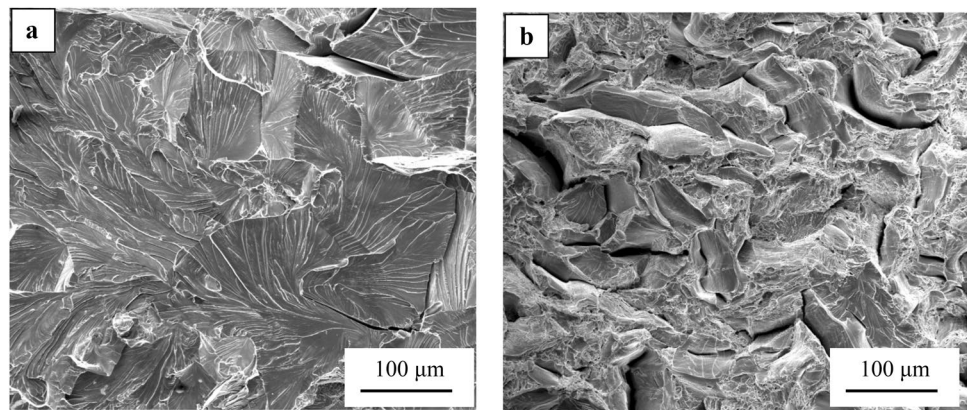
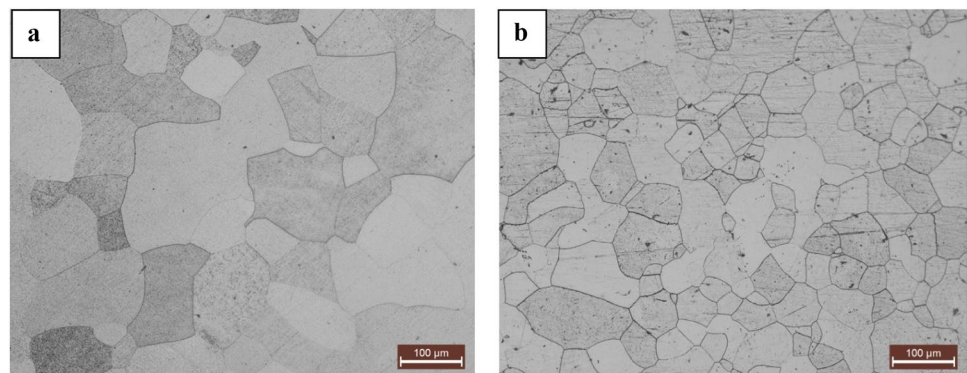
Fig. 5 Fractographs of tensile tested samples processed under warm rolled and annealed conditions: (a) Nb–W alloy and (b) Nb–W–Zr alloy**Fig. 6** Optical micrographs of alloys processed under cold rolled and annealed conditions: (a) Nb–W alloy and (b) Nb–W–Zr alloy

Fig. 7 SEM micrographs of alloys processed under cold rolled and annealed conditions: (a) Nb–W alloy and (b) Nb–W–Zr alloy

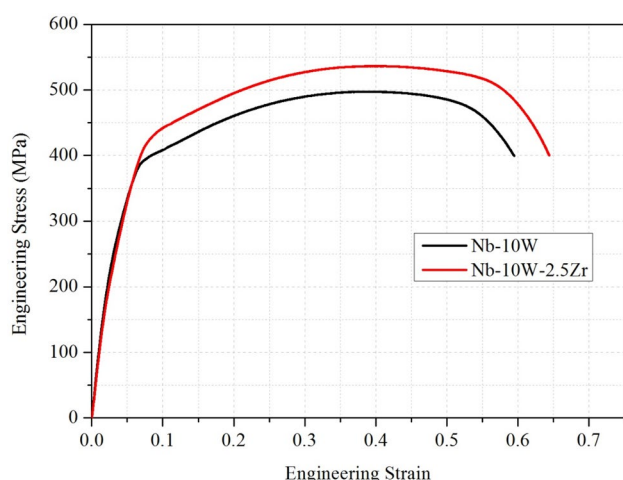
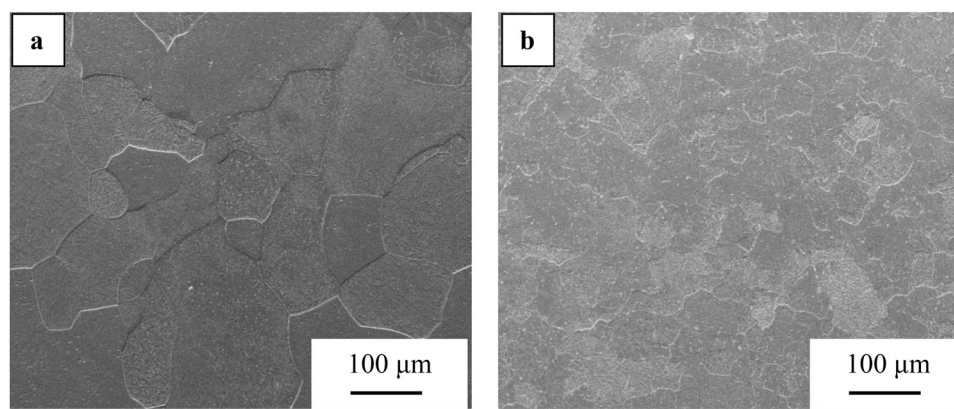


Fig. 8 Plot showing tensile curves of cold rolled and annealed Nb–W and Nb–W–Zr alloys

Table 3 Tensile properties of Nb–W and Nb–W–Zr alloys processed under cold rolled and annealed condition

Alloys	Mechanical properties		
	YS (MPa)	UTS (MPa)	Elongation (%)
Nb–W	382 ± 3.5	492 ± 3	37 ± 1
Nb–W–Zr	413 ± 2.5	541 ± 2	38 ± 1

As pure Nb has low strength, it is alloyed with W to increase its strength. The increase of strength of Nb by W addition is mainly due to the combined effect of size and modulus misfit associated with Nb and W atoms ($r_{\text{Nb}} = 145$ pm and $r_{\text{W}} = 136$ pm; $G_{\text{Nb}} = 38$ GPa and $G_{\text{W}} = 160$ GPa). However, the addition of W content is limited to 11% to maintain a reasonable plasticity in Nb alloy [25]. The potency of strengthening of Nb by W addition is the same for both alloys as W concentration is the same in both alloys. The additional strength observed in the Nb–W–Zr alloy as compared to the Nb–W alloy

might have come from the grain boundary strengthening and solid solution strengthening of Nb–W by Zr. In addition, fine scale Zr-rich precipitates may be expected to be formed in Nb–W–Zr alloy. According to Nb–Zr phase diagram, the solid solubility of Zr in Nb is very low at room temperature because of their differences in crystal structure and atomic size ($r_{\text{Nb}} = 145$ pm and $r_{\text{Zr}} = 160$ pm). It may be inferred from the Nb–Zr binary phase diagram that it is a completely immiscible system. Further it has been reported that the enthalpy of mixing of Nb and Zr is positive ($\Delta H_{\text{mix}} = 17 \text{ kJ mol}^{-1}$) [26]. Thus, Zr precipitates are likely to be present in Nb–W–Zr alloy. The evidences of Zr precipitates in Nb matrix have also been reported for Nb–Zr binary alloys [27]. In the present study, the presence of submicron sized Zr precipitates may be partially responsible for obtaining higher strength values in Nb–W–Zr alloy. However, the kinetics of precipitation of Zr in Nb-based alloys is still not fully understood. The EBSD analysis was carried out on cold rolled and annealed Nb–W–Zr alloy in order to qualitatively make out whether any ZrO_2 particles and Zr precipitates are present or not. The EBSD phase mapping and inverse pole figure (IPF) orientation image of Nb–W–Zr alloy is illustrated in Fig. 9 and Fig. 10, respectively. It can be seen that the alloy exhibits a three distinct kinds of particles in Nb matrix. They are qualitatively identified as hexagonal Zr, tetragonal ZrO_2 and monoclinic ZrO_2 . The corresponding lattice parameters of them have also been measured. The lattice parameter of hexagonal Zr is $a = b = 3.23$ Å and $c = 5.15$ Å, the lattice parameter of tetragonal ZrO_2 is $a = b = 3.64$ Å, $c = 5.27$ Å and the lattice parameter of monoclinic ZrO_2 is $a = 5.14$ Å, $b = 5.20$ Å, $c = 5.31$ Å. These lattice parameters values are comparable to the lattice parameters reported for these phases in the literature [28]. However, a further in-depth investigation of these precipitates using a transmission electron microscopy (TEM) is essential to quantitatively pinpoint whether any Zr precipitates are present or not in Nb–W–Zr alloy. Further the TEM analysis would also be helpful to predict whether the Zr

Fig. 9 EBSD (a) band contrast image and (b) EBSD phase mapping of cold rolled and annealed Nb–W–Zr alloy. (Red colour: Nb–W matrix, Blue colour: Zr precipitates, Yellow colour: Tetragonal-ZrO₂, Green colour-Monoclinic ZrO₂)

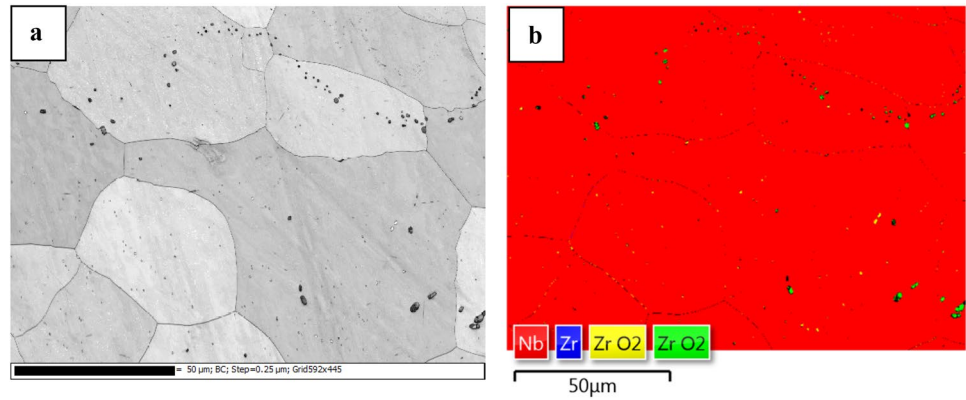


Fig. 10 EBSD IPF orientation images showing the presence of Tetragonal-ZrO₂ and Monoclinic ZrO₂

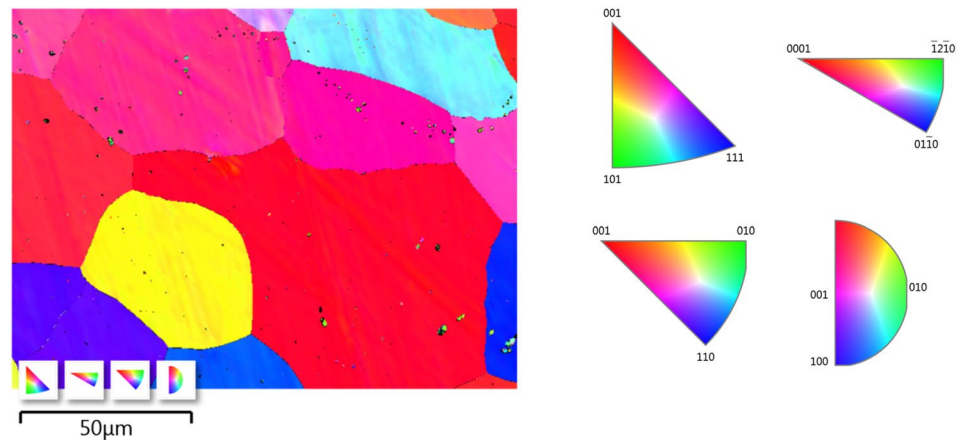
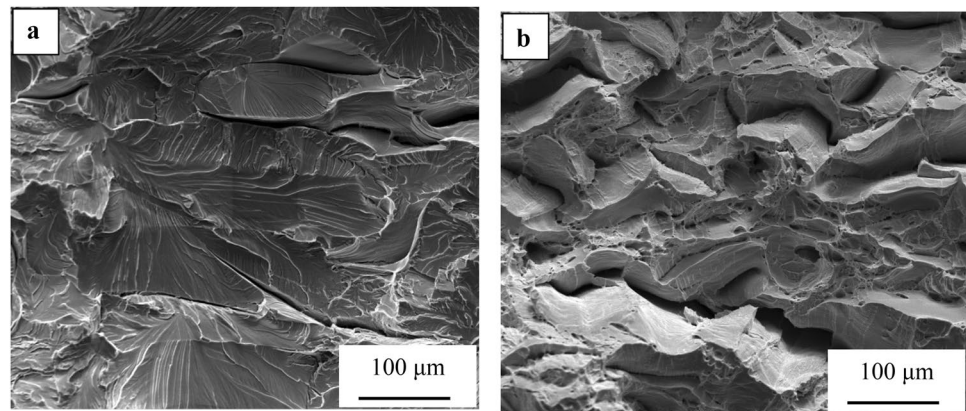


Fig. 11 Fractographs of tensile tested samples processed under cold rolled and annealed conditions: (a) Nb–W alloy and (b) Nb–W–Zr alloy



precipitates have any coherency with Nb–W matrix or not. Figure 11 displays the fractography of the cold rolled and annealed alloys. The Nb–W alloy displays a completely transgranular cleavage fracture. This is mainly attributed to the presence of relatively coarse grains in Nb–W alloy. The addition of Zr results in the formation of a relatively fine grain in Nb–W–Zr alloy due to the presence of ZrO₂ particles, which act as a grain refiner. As a result, the fracture mode of the alloy is significantly changed from

a completely transgranular cleavage for Nb–W alloy to a mixed mode of fracture consisting of both dimple and cleavage for Nb–W–Zr alloy. It shows that Zr addition not only enhances the strength but also changes the fracture behaviour of the alloy. It is understood from fractography that the processing method does not have an influence on the fracture mode of Nb–W and Nb–W–Zr alloys.

4 Conclusions

Nb–10W and Nb–10W–2.5Zr ingots were produced through EBM process using the raw materials in the form of powders. Subsequently they were subjected to thermomechanical processing to produce plates and sheets. The following conclusions are drawn based on the microstructure and tensile properties.

1. The electron beam melted ingots of both Nb–W and Nb–W–Zr alloy exhibit a single-phase microstructure. The alloys possess the required chemical composition with low levels of interstitial impurities.
2. Nb–W–Zr alloy exhibits relatively fine grains structure as compared to Nb–W alloy.
3. Nb–W alloy displays a sharp yield point during the tensile test in warm rolled and annealed condition. The yield point phenomenon is not observed for Nb–W–Zr alloy tested under cold rolled and annealed condition. Nb–W–Zr alloy does not show any yield point phenomenon during tensile test.
4. The strength of Nb–W–Zr alloy is higher than that of Nb–W alloy in both warm rolled and annealed as well as cold rolled and annealed conditions.
5. Nb–W alloy displays a completely transgranular cleavage fracture mode, while Nb–W–Zr alloy shows a mixed fracture mode.

Acknowledgements The authors are grateful to Defence Research and Development Organisation, Ministry of Defence, New Delhi, for the financial support in carrying out this research work under DMR–295 project. The authors wish to thank Dr. R. Balamuralikrishnan, Director, DMRL for his keen interest and encouragement. The authors also would like to thank officers and staff of AMTG, MWG, SFAG, MBG and EMG for giving their technical support to carry out this work

References

1. Sandim H R Z, and Padilha A K, *Key Engg Mater* **189–191** (2001) 296.
2. Sankar M, Satish R Y, and Baligidad R G, *Trans IIM* **62** (2), (2009) 135.
3. Sankar M, Baligidad R G, Satyanarayana D V V, and Gokhale A A, *Mater Sci Eng A* **574** (2013) 104.
4. Valdevit L, Vermaak N, Zok F W, and Evans A G, *J Appl Mech* **75** (2008) 1.
5. Wilcox BA, Chapter 1: in *Refractory Metal Alloys Metallurgy and Technology* edited by Machlin I, Begley R T, E. D. Weisert, Plenum Press, New York (1968) 1
6. Slining J R, and Koss D A, *Metall Trans* **4** (1973) 1261.
7. Sankar M, Baligidad R G, and Gokhale A A, *Mater Sci Engg A* **569** (2013) 132.
8. Loomis B A, and Gerber B, *Scr Metall* **4** (11), (1970) 921.
9. Rointan F B, *Vac*, Reinhold Pub.Co, New York, sMetall (1958), p 162.
10. Moss A R, and Richards D T, *J Less common Met* **2** (6), (1960) 405.
11. Cook M, and Swainson E, *J Inst Met* **87** (1959) 161.
12. StJohn D H, Prasad A, Easton M A, and Qian M, *Metall Mater Trans A* **46** (2015) 4868.
13. Banu A, Ali M Y, and Rahman M A, *Int J Adv Manuf Technol* **75** (2014) 257.
14. Chevallier J, Gremillard L, Virkar A V, and Clarke D R, *J Am Ceram Soc* **92** (9), (2009) 1901.
15. Nistico R, *J Aust Ceram Soc* **57** (2021) 225.
16. Candolpho K C G, and Nunes C A, *Int J Refract Met Hard Mater* **24** (2006) 413.
17. Fromm E, and Jehn H, *Vacuum* **19** (4), (1969) 191.
18. Roth A, Vacuum Technology, Amsterdam, the Netherlands, North-Holland Publishing Company, (1982) p 157.
19. Upadhyaya G S, *Bull Mater Sci* **28** (2005) 305.
20. Gao S, Chen M, Chen S, Kamikawa N, Shibata A, and Tsuji N, *Mater Trans* **55** (1), (2014) 73.
21. Yang P J, Li Q J, Han W Z, Li J, and Ma E, *Acta Mater* **179** (2019) 107.
22. Diaz A E, and Reed-hill R E, *Scr Mater* **13** (6), (1979) 491.
23. Colas D, Finot E, Flouriot S, Forest S, Maziere M, and Paris T, *Mater Sci Eng A* **615** (2014) 283.
24. Hall E O, *Yield point phenomena in metals and alloys*, Springer, US (1970), p 125.
25. Scott M H, and Knowlson P M, *J Less Common Met* **5** (3), (1963) 205.
26. Pineau A, Benzerger A A, and Pardoën T, *Acta Mater* **101** (2016) 4241.
27. Al-Aqeeli N, Hussein M A, and Suryanarayana C, *Adv Powder Metall* **26** (2015) 385.
28. Morton N, James B W, Wostenholm G H, and Nichols R J, *J. Phys F Met Phys* **5** (1975) 85.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.